

Laptop Sales Analysis Project Report

1.Introduction

- The laptop industry offers a wide range of products with different prices and hardware configurations. Analyzing laptop data helps understand market trends, pricing patterns, and the impact of specifications on price. This project performs Exploratory Data Analysis (EDA) on a laptop dataset to identify useful insights using Python. The analysis supports better business decisions related to pricing, product positioning, and market trends.

2. Business Problem Statement

- With many laptop brands and models available, it is difficult for customers to compare products and for businesses to develop effective pricing strategies. This project analyzes laptop specifications and prices to identify the key factors affecting pricing, compare brands, and provide insights that help businesses make informed decisions.
- This project addresses these challenges by analyzing a laptop dataset to identify factors affecting laptop prices, compare brands based on hardware specifications, discover pricing patterns, and generate actionable business insights

that can support better product positioning and business decision-making.

3. Business Objectives

- The primary objective of this project is to analyze laptop market data and generate meaningful insights that support business decision-making. The specific objectives are:
 - To analyze the distribution of laptop prices across different brands.
 - To identify the relationship between hardware specifications (RAM, Processor, Storage, GPU, and Operating System) and laptop price.
 - To compare laptop brands based on pricing and technical specifications.
 - To identify the most common hardware configurations available in the market.
 - To categorize laptops into budget, mid-range, and premium segments.
 - To examine how processor generation and graphics cards influence pricing.
 - To identify premium brands and brands offering the best value for money.
 - To detect unusual or outlier laptop prices that may require further investigation.

4. Dataset Description

- The dataset contains information about different laptop models from various brands.
- It includes **27,456 records** and **18 columns**.
- The dataset contains features such as **Brand, Model Name, Price, CPU, GPU, RAM, Storage, Operating System, CPU Generation, and Usage Purpose**.
- It is used to analyze laptop pricing, hardware specifications, and market trends.
- The dataset helps compare brands, identify pricing patterns, and generate business insights through Exploratory Data Analysis (EDA).

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	brand	name	price	cpu	gpu	storage	ram	os	extract_cp	cpu_ryzer	intel_core	cpu_gen	cpu_priori	os_priori	gpu_prior	ram_prior	storage_p	usage_purpose	
2	ASUS	TUF A16 F	43146	amd ryzer	rtx 4050	512 GB	16 GB	FreeDOS	amd ryzer	7		7445h		1	4	5	7	Gaming / Design	
3	ASUS	Vivobook	29149	intel core 5	120u	512 GB	16 GB	Windows	intel core			120u		2	3		5	7	Office
4	ASUS	Vivobook	9999	intel celeron	n4500	128 GB	4 GB	Windows	11 Home			4500		1	3		1	2	Daily
5	ASUS	TUF Gamii	42449	amd ryzer	rtx 4050	512 GB	16 GB	FreeDOS	amd ryzer	7		7445h		1	4	5	7	Gaming / Design	
6	ASUS	TUF F16 F	48799.1	intel core	rtx 5050	512 GB	16 GB	FreeDOS	intel core			5 13450hx		3	1	4	5	7	Gaming / Design
7	ASUS	Vivobook	21099	amd ryzen5	7520u	512 GB	8 GB	FreeDOS	amd ryzer	5		7520u			1		3	7	Daily
8	ASUS	TUF GAMII	45799	intel core	rtx 5050	512 GB	16 GB	FreeDOS	intel core			5 13450hx		3	1	4	5	7	Gaming / Design
9	ASUS	Vivobook	27599	intel core 5	120u	512 GB	16 GB	Windows	intel core			120u		2	4		5	7	Office
10	ASUS	15.6-inch,	19719	amd ryzen5	7520u				amd ryzer	5		7520u							Daily

• PHASE 1 : Data Import & Exploration

DATA IMPORT :

The laptop dataset was imported into a Pandas DataFrame using the `read_csv()` function for further analysis.

```
# Import Required Libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

DATA LOADING :

The laptop dataset was successfully loaded into a Pandas DataFrame, allowing it to be explored, cleaned, and analyzed.

```
# Load the Dataset

df = pd.read_csv("laptop_dataset.csv")

print("Dataset Loaded Successfully!")

Dataset Loaded Successfully!
```

DATA EXPLORATION :

Basic structural and statistical checks are performed:

- **df.info()** to check data types, null values, and column structure

```
# Dataset Information

df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27456 entries, 0 to 27455
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   brand                 27456 non-null  object
1   name                 27456 non-null  object
2   price                27456 non-null  float64
3   cpu                  27456 non-null  object
4   gpu                  16975 non-null  object
5   storage              27400 non-null  object
6   ram                  27419 non-null  object
7   os                   27294 non-null  object
8   extract_cpu_brand    27444 non-null  object
9   cpu_ryzen_series     3749 non-null   float64
10  intel_core_series     12628 non-null  float64
11  cpu_gen               27252 non-null  object
12  cpu_priority          23702 non-null  float64
13  os_priority           27294 non-null  float64
14  gpu_priority          16975 non-null  float64
15  ram_priority          27419 non-null  float64
16  storage_priority      27400 non-null  float64
17  usage_purpose           27456 non-null  object
dtypes: float64(8), object(10)
memory usage: 3.8+ MB
```

• PHASE 2 : Data Cleaning & Preprocessing

STEP 1 : CHECK MISSING VALUES.

Identified missing values in each column using `isnull().sum()` and calculated their percentage.

```
# Check Missing Values

missing_values = df.isnull().sum()

missing_df = pd.DataFrame({
    "Missing Values": missing_values,
    "Percentage (%)": round((missing_values / len(df)) * 100, 2)
})

missing_df
```

	Missing Values	Percentage (%)
brand	0	0.00
name	0	0.00
price	0	0.00
cpu	0	0.00
gpu	10481	38.17
storage	56	0.20
ram	37	0.13
os	162	0.59
extract_cpu_brand	12	0.04
cpu_ryzen_series	23707	86.35
intel_core_series	14828	54.01
cpu_gen	204	0.74
cpu_priority	3754	13.67
os_priority	162	0.59
gpu_priority	10481	38.17
ram_priority	37	0.13
storage_priority	56	0.20
usage_purpose	0	0.00

STEP 2 : CHECK DUPLICATE RECORDS.

Checked for duplicate rows using the **uplicated()** function.

```
# Check Duplicate Rows

duplicate_rows = df.duplicated().sum()

print("Duplicate Rows:", duplicate_rows)

Duplicate Rows: 2846
```

STEP 3 : REMOVE DUPLICATE RECORDS.

Removed duplicate records using **drop_duplicates()** to improve data quality.

```
# Remove Duplicate Rows

df = df.drop_duplicates()

print("Dataset Shape After Removing Duplicates:", df.shape)

Dataset Shape After Removing Duplicates: (24610, 18)
```

STEP 4 : CHECK DATA TYPES.

Reviewed the data types of all columns using **dtypes** to ensure they were appropriate for analysis.

```
# Check Data Types
```

```
df.dtypes
```

```
brand          object
name           object
price          float64
cpu            object
gpu            object
storage        object
ram            object
os             object
extract_cpu_brand  object
cpu_ryzen_series float64
intel_core_series float64
cpu_gen        object
cpu_priority    float64
os_priority     float64
gpu_priority    float64
ram_priority    float64
storage_priority float64
usage_purpose     object
dtype: object
```

STEP 5 : CHECK MISSING VALUES.

Rechecked the dataset to confirm the missing values status before analysis.

```
# Verify Missing Values
```

```
df.isnull().sum()
```

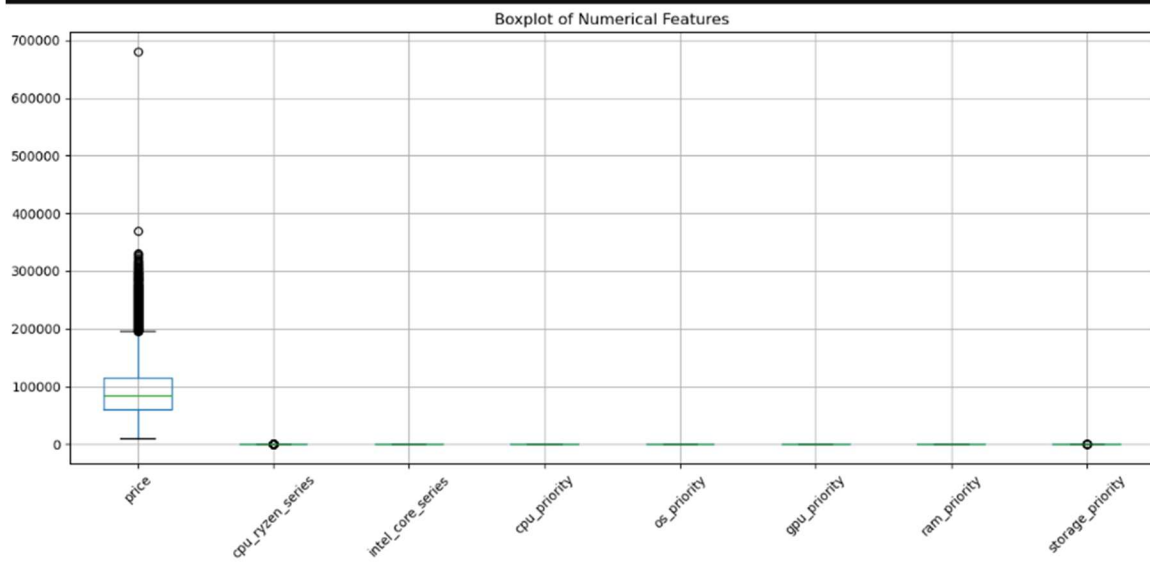
```
brand          0
name           0
price          0
cpu            0
gpu           9127
storage        50
ram           32
os            143
extract_cpu_brand  12
cpu_ryzen_series 21259
intel_core_series 13241
cpu_gen        193
cpu_priority    3356
os_priority     143
gpu_priority    9127
ram_priority    32
storage_priority 50
usage_purpose     0
dtype: int64
```

STEP 6 : DETECT OUTLIERS.

Used boxplots to identify outliers in numerical columns before performing Exploratory Data Analysis (EDA).

```
numerical_cols = df.select_dtypes(include=["int64", "float64"]).columns

plt.figure(figsize=(15,6))
df[numerical_cols].boxplot()
plt.xticks(rotation=45)
plt.title("Boxplot of Numerical Features")
plt.show()
```



STEP 7 : Final Dataset Information.

Verified the final dataset structure using `df.info()` to ensure it was ready for analysis.

```
# Final Dataset Information

df.info()

<class 'pandas.core.frame.DataFrame'>
Index: 24610 entries, 0 to 27454
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   brand                  24610 non-null  object
1   name                   24610 non-null  object
2   price                  24610 non-null  float64
3   cpu                    24610 non-null  object
4   gpu                    15483 non-null  object
5   storage                24560 non-null  object
6   ram                    24578 non-null  object
7   os                     24467 non-null  object
8   extract_cpu_brand     24598 non-null  object
9   cpu_ryzen_series       3351 non-null   float64
10  intel_core_series      11369 non-null  float64
11  cpu_gen                24417 non-null  object
12  cpu_priority           21254 non-null  float64
13  os_priority            24467 non-null  float64
14  gpu_priority           15483 non-null  float64
15  ram_priority           24578 non-null  float64
16  storage_priority       24560 non-null  float64
17  usage_purpose            24610 non-null  object
dtypes: float64(8), object(10)
memory usage: 3.6+ MB
```

STEP 8 : Save Cleaned Dataset

Saved the cleaned dataset as **Laptop_Cleaned_Dataset.csv** using the `to_csv()` function.

```
# Save Cleaned Dataset

df.to_csv("Laptop_Cleaned_Dataset.csv", index=False)

print("Cleaned Dataset Saved Successfully!")

Cleaned Dataset Saved Successfully!
```

• PHASE 3: Exploratory Data Analysis (EDA)

Q1. What is the distribution of laptop brands?

This analysis shows the number of laptop models available for each brand, helping identify the brands with the highest market presence in the dataset.

```
Q1. What is the distribution of laptop brands?
```

```
brand_counts = df["brand"].value_counts()
```

```
brand_counts
```

brand	
MSI	4509
HP	4230
Dell	3999
LENOVO	3782
ACER	3650
ASUS	3576
CASPER	714
MONSTER	99
Huawei	30
Apple	21

```
Name: count, dtype: int64
```

Q2. Which are the Top 10 laptop brands?

This analysis identifies the top 10 brands with the highest number of laptop models, providing a comparison of the leading brands in the dataset.

Q2. Which are the Top 10 laptop brands?

```
df["brand"].value_counts().head(10)
```

```
brand
MSI      4509
HP       4230
Dell     3999
LENOVO   3782
ACER     3650
ASUS     3576
CASPER   714
MONSTER  99
Huawei    30
Apple    21
Name: count, dtype: int64
```

Q3. What is the average price of laptops by brand?

This analysis compares the average laptop price across different brands, helping identify premium and budget-friendly brands.

```
df.groupby("brand")["price"].mean().sort_values(ascending=False)
```

```
brand
MSI      117026.863781
ACER     111530.397526
HP       89222.412608
LENOVO   84277.128458
ASUS     77181.641725
Dell     77059.135714
MONSTER  76100.198889
Apple    69080.049524
CASPER   62349.017605
Huawei    32042.157000
Name: price, dtype: float64
```

Q4. Which are the Top 10 most expensive laptops?

This analysis identifies the top 10 highest-priced laptops in the dataset, highlighting premium models with high-end specifications.

Q4. Which are the Top 10 most expensive laptops?

```
df[["brand","name","price"]].sort_values(by="price", ascending=False).head(10)
```

	brand	name	price
4519	LENOVO	Legion 9 18IAX10 Ultra 9 275HX 192GB 16TBSSD W...	679634.50
17925	Dell	Alienware 18 AA18250 Ultra 9 275HX 128GB 4TBSS...	369764.98
18245	MSI	NB TITAN 18 HX AI A2XWJG-620TR ULTRA 9 285HX 6...	329999.00
18252	MSI	TITAN 18 HX AI A2XWJG-217TR 24GB RTX5090 Intel...	329699.00
27322	ACER	Predator PH18-73 Ultra 9 275HX 96GB 8tb W11PRO...	325511.50
27395	ACER	Predator PH18-73 Ultra 9 275HX 96GB 8tb W11Hom...	322145.50
27217	ACER	Predator PH18-73 Ultra 9 275HX 192GB 8tb W11PR...	319274.50
27423	ACER	Predator PH18-73 Ultra 9 275HX 96GB 6tb W11PRO...	315710.50
27373	ACER	Predator PH18-73 Ultra 9 275HX 192GB 8tb W11Ho...	315512.50
27422	ACER	Predator PH18-73 Ultra 9 275HX 96GB 6tb W11Hom...	312443.50

Q5. Which are the 10 cheapest laptops?

This analysis identifies the 10 lowest-priced laptops, helping understand the budget segment and affordable options available in the market.

Q5. Which are the 10 cheapest laptops?

```
df[["brand","name","price"]].sort_values(by="price").head(10)
```

	brand	name	price
2	ASUS	Vivobook Go Intel Celeron N4500 4GB 128GB SSD ...	9999.0
23620	ACER	Aspire Lite AL15-32P Intel Celeron-N4500 4GB 1...	10199.0
65	ASUS	Vivobook Go 15 E510Ka-Ej483W Intel Celeron N45...	11500.0
4418	LENOVO	IdeaPad 1 Intel Celeron N4020 4GB 128GB SSD Wi...	11999.0
22906	CASPER	Nirvana Intel Celeron N4020 4GB RAM 120GB SSD ...	12149.0
55	ASUS	E510KA-EJ881W Vivobook Go 15/Intel Celeron N45...	12479.0
9	ASUS	E510Ma-Ej592W Intel Celeron N4020 4GB RAM 128G...	12480.0
22914	CASPER	Nirvana Celeron N4020-4Gb-120Gb Ssd-15.6inc-W11	12799.0
4446	LENOVO	Ideapad1 Celeron N4120 4gb Ram 128gb 15.6" Win...	12999.0
23621	ACER	Aspire Lite AL15-32P Intel Celeron N4500 8GB 2...	12999.0

Q6. Which operating system is most popular?

This analysis identifies the most frequently used operating system among laptops, helping understand the market preference for different operating systems.

```
Q6. Which operating system is most popular?

df["os"].value_counts()

os
FreeDOS      8097
Windows 11 Pro 7787
Windows 11 Home 7306
Windows 10 Pro  499
Windows 10 Home 473
Linux         278
macOS         21
Windows        6
Name: count, dtype: int64
```

Q7. What is the average laptop price for each operating system?

This analysis compares the average price of laptops based on their operating system, showing how OS influences laptop pricing.

```
Q7. What is the average laptop price for each operating system?

df.groupby("os")["price"].mean().sort_values(ascending=False)

os
Windows 10 Pro    96619.172345
Windows 10 Home   96338.410148
Windows 11 Pro    95033.021558
Windows 11 Home   94848.183312
FreeDOS           88158.326112
macOS             69080.049524
Linux             62354.526835
Windows           54425.700000
Name: price, dtype: float64
```

Q8. Which CPU brand is used the most?

This analysis identifies the most commonly used CPU brand in the dataset, helping understand the processor preferences of laptop manufacturers.

Q8. Which CPU brand is used the most?

```
df["extract_cpu_brand"].value_counts()
```

```
extract_cpu_brand
intel core      19989
amd ryzen       4580
apple           24
snapdragon       5
Name: count, dtype: int64
```

Q9. What is the average price based on CPU brand?

This analysis compares the average laptop price for different CPU brands, showing how processor type influences pricing.

Q9. What is the average price based on CPU brand?

```
df.groupby("extract_cpu_brand")["price"].mean().sort_values(ascending=False)
```

```
extract_cpu_brand
amd ryzen      93202.585439
intel core     92089.828161
apple          64065.751667
snapdragon     38985.400000
Name: price, dtype: float64
```

Q10. Which GPU is most commonly used?

This analysis identifies the most frequently used GPU models, helping understand the popularity of different graphics cards in the laptop market.

Q10. Which GPU is most commonly used?

```
df["gpu"].value_counts().head(15)
```

```
gpu
rtx 5060      4574
rtx 5070      2354
rtx 4050      1680
rtx 5070ti    1548
rtx 4060      1427
rtx 5050      1064
rtx 3050       861
rtx 4070       619
rtx 5080       286
intel arc graphics  229
rtx 2050       181
intel uhd graphics  175
rtx a500       150
intel iris xe     74
rtx a1000       61
Name: count, dtype: int64
```

Q11. What is the average price by RAM size?

This analysis compares the average laptop price across different RAM sizes, showing how higher RAM configurations generally increase laptop prices.

Q11. What is the average price by RAM size?

```
df.groupby("ram")["price"].mean().sort_values()
```

```
ram
4 GB      19811.894737
12 GB     44752.936278
6 GB      47099.400000
20 GB     48598.551795
36 GB     54730.713017
8 GB      62856.407680
16 GB     66849.490385
24 GB     80685.513969
32 GB     86636.588455
40 GB     89172.403824
48 GB     102454.797518
64 GB     110799.076063
18 GB     119932.666667
80 GB     148116.528682
96 GB     167698.897126
128 GB    205157.447013
Name: price, dtype: float64
```

Q12. Which RAM configuration is most popular?

This analysis identifies the most commonly used RAM configuration among laptops, helping understand the standard memory capacity preferred in the market.

```
Q12. Which RAM configuration is most popular?

df["ram"].value_counts()

ram
16 GB    4259
32 GB    3973
64 GB    3193
24 GB    3079
48 GB    2756
40 GB    2722
8 GB     1879
96 GB    1089
80 GB     827
128 GB    318
12 GB     223
20 GB     117
36 GB     116
4 GB        19
6 GB         5
18 GB         3
Name: count, dtype: int64
```

Q13. Which storage option is most common?

This analysis identifies the most frequently available storage capacities, helping understand the storage configurations commonly offered in laptops.

```
Q13. Which storage option is most common?

df["storage"].value_counts().head(10)

storage
1 TB      6036
512 GB    5355
2 TB      5084
4 TB      4467
256 GB    3006
8 TB       202
480 GB     178
6 TB       171
500 GB      31
128 GB      24
Name: count, dtype: int64
```

Q14. Which storage type has the highest average price?

This analysis compares the average price of laptops with different storage capacities, showing how storage size affects laptop pricing.

Q14. Which storage type has the highest average price?

```
df.groupby("storage")["price"].mean().sort_values(ascending=False)
```

```
storage
128 GB    221948.610417
8 TB      162714.860594
6 TB      151703.757310
4 TB      111542.157634
2 TB      94563.234839
256 GB    90021.493739
1 TB      85877.149942
512 GB    78973.952063
500 GB    53481.380968
480 GB    47368.191011
250 GB    20499.000000
120 GB    13127.000000
Name: price, dtype: float64
```

Q15. Which usage purpose has the most laptops?

This analysis identifies the most common laptop usage category, helping understand the primary purpose for which laptops are designed.

Q15. Which usage purpose has the most laptops?

```
df["usage_purpose"].value_counts()
```

```
usage_purpose
Gaming / Design    14924
Office              8698
Daily               988
Name: count, dtype: int64
```

Q16. What is the average price for each usage purpose?

This analysis compares the average price of laptops across different usage purposes, highlighting the price differences between Gaming, Office, and Daily-use laptops.

Q16. What is the average price for each usage purpose?

```
df.groupby("usage_purpose")["price"].mean().sort_values(ascending=False)
```

```
usage_purpose
Gaming / Design    109619.757716
Office             68280.158541
Daily              40146.761457
Name: price, dtype: float64
```

Q17. Which Intel Core series is most common?

This analysis identifies the most frequently used Intel Core series, helping understand the popularity of different Intel processors in the laptop market.

Q17. Which Intel Core series is most common?

```
df["intel_core_series"].value_counts()
```

```
intel_core_series
7.0    5991
5.0    4145
9.0    1193
3.0     40
Name: count, dtype: int64
```

Q18. Which CPU generation appears the most?

This analysis identifies the most commonly used CPU generations, helping understand the latest processor generations available in the laptop market.

```
Q18. Which CPU generation appears the most?  
  
df["cpu_gen"].value_counts().sort_index()  
  
cpu_gen  
1005      1  
100u     30  
10310u    1  
1035      2  
1115      3  
...  
8940hx    342  
8945       5  
8945h     25  
9955       2  
9955hx    71  
Name: count, Length: 135, dtype: int64
```

Q19. Which brand has the greatest variety of laptop models?

This analysis compares the number of unique laptop models offered by each brand, highlighting brands with the widest product range.

```
Q19. Which brand has the greatest variety of laptop models?  
  
df.groupby("brand")["name"].nunique().sort_values(ascending=False)  
  
brand  
MSI      4508  
HP       4229  
Dell     3989  
LENOVO   3777  
ACER     3650  
ASUS     3556  
CASPER   714  
MONSTER  99  
Huawei    30  
Apple    21  
Name: name, dtype: int64
```

Q20. Which brand has the highest average laptop price?

This analysis compares the average laptop price across different brands, helping identify the brands positioned in the premium laptop segment.

Q20. Which brand has the highest average laptop price?

```
df.groupby("brand")["price"].mean().sort_values(ascending=False).head(10)
```

```
brand
MSI      117026.863781
ACER     111530.397526
HP       89222.412608
LENOVO   84277.128458
ASUS     77181.641725
Dell     77059.135714
MONSTER  76100.198889
Apple    69080.049524
CASPER   62349.017605
Huawei    32042.157000
Name: price, dtype: float64
```

- **PHASE 4 : Data Visualization**

Chart 1: Top 10 Laptop Brands (Bar Chart)

Interpretation: The bar chart compares the number of laptop models offered by the top 10 brands. It helps identify the brands with the strongest market presence.

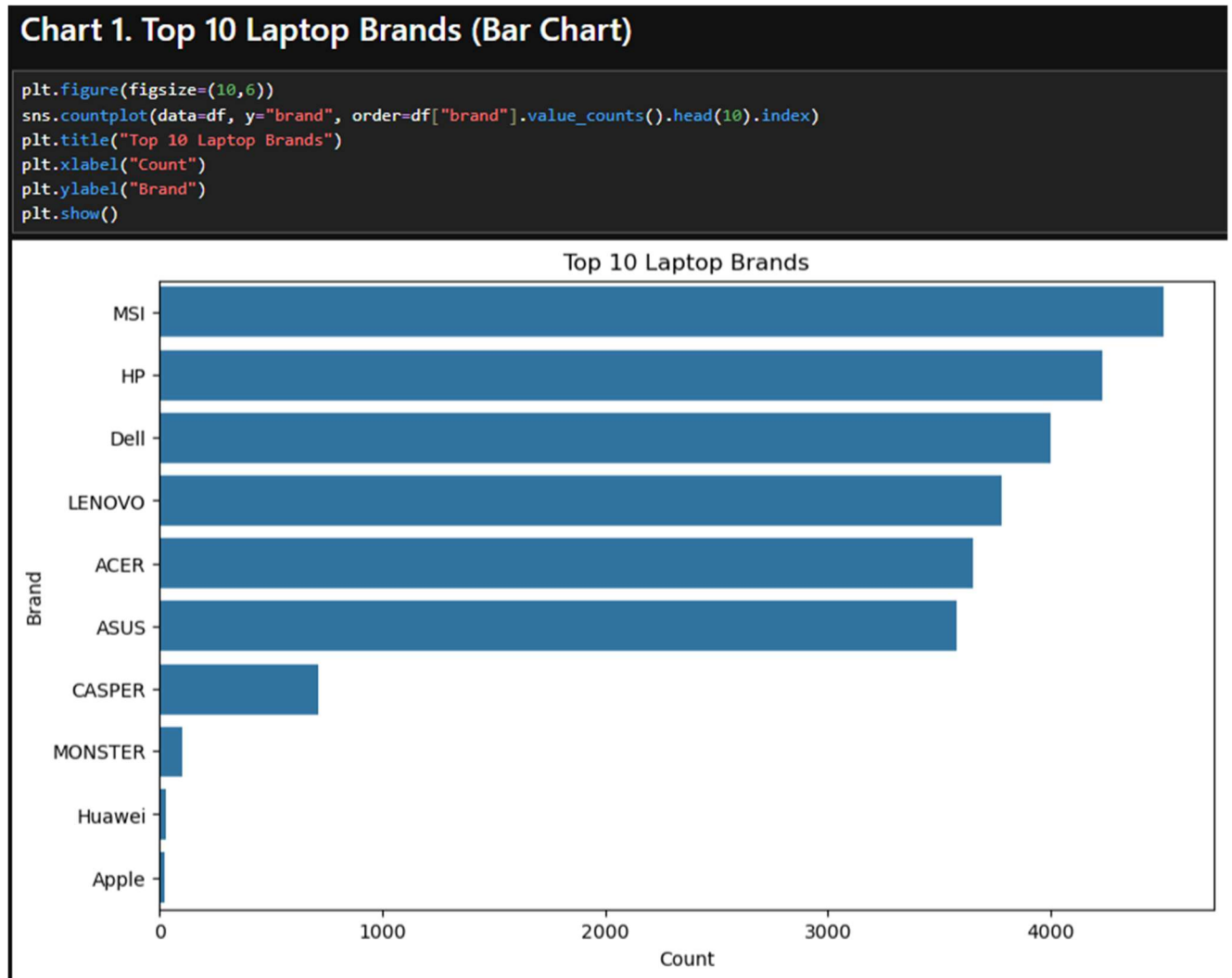


Chart 2: Laptop Price Distribution (Histogram)

Interpretation: The histogram shows how laptop prices are distributed across different price ranges. It helps identify the most common price segments and overall pricing trends.

Chart 2. Laptop Price Distribution (Histogram)

```
plt.figure(figsize=(10,6))
sns.histplot(df["price"], bins=30, kde=True)
plt.title("Distribution of Laptop Prices")
plt.xlabel("Price")
plt.ylabel("Frequency")
plt.show()
```

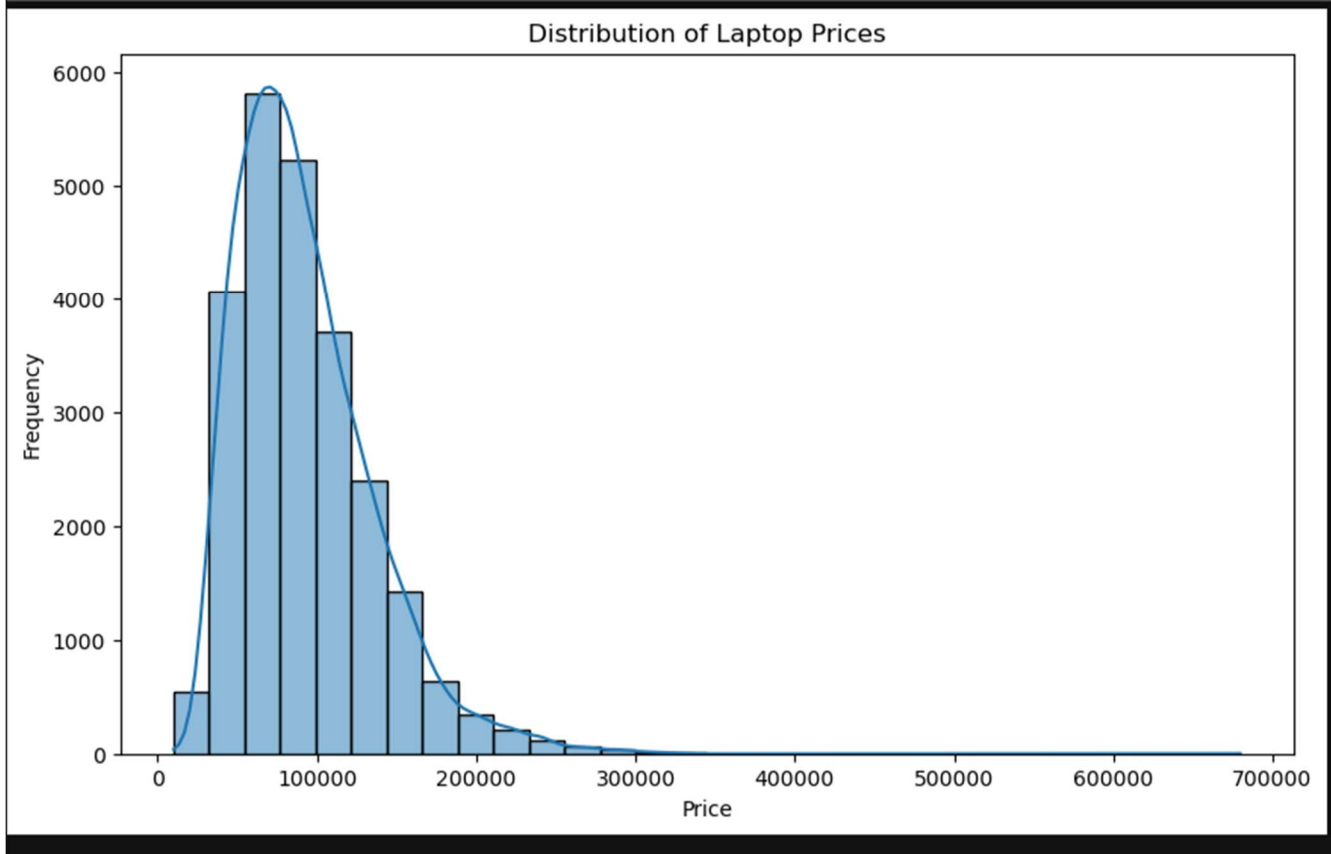


Chart 3: Brand vs Operating System (Heatmap)

Interpretation: The heatmap shows the relationship between laptop brands and operating systems, helping identify which operating systems are most commonly offered by each brand.

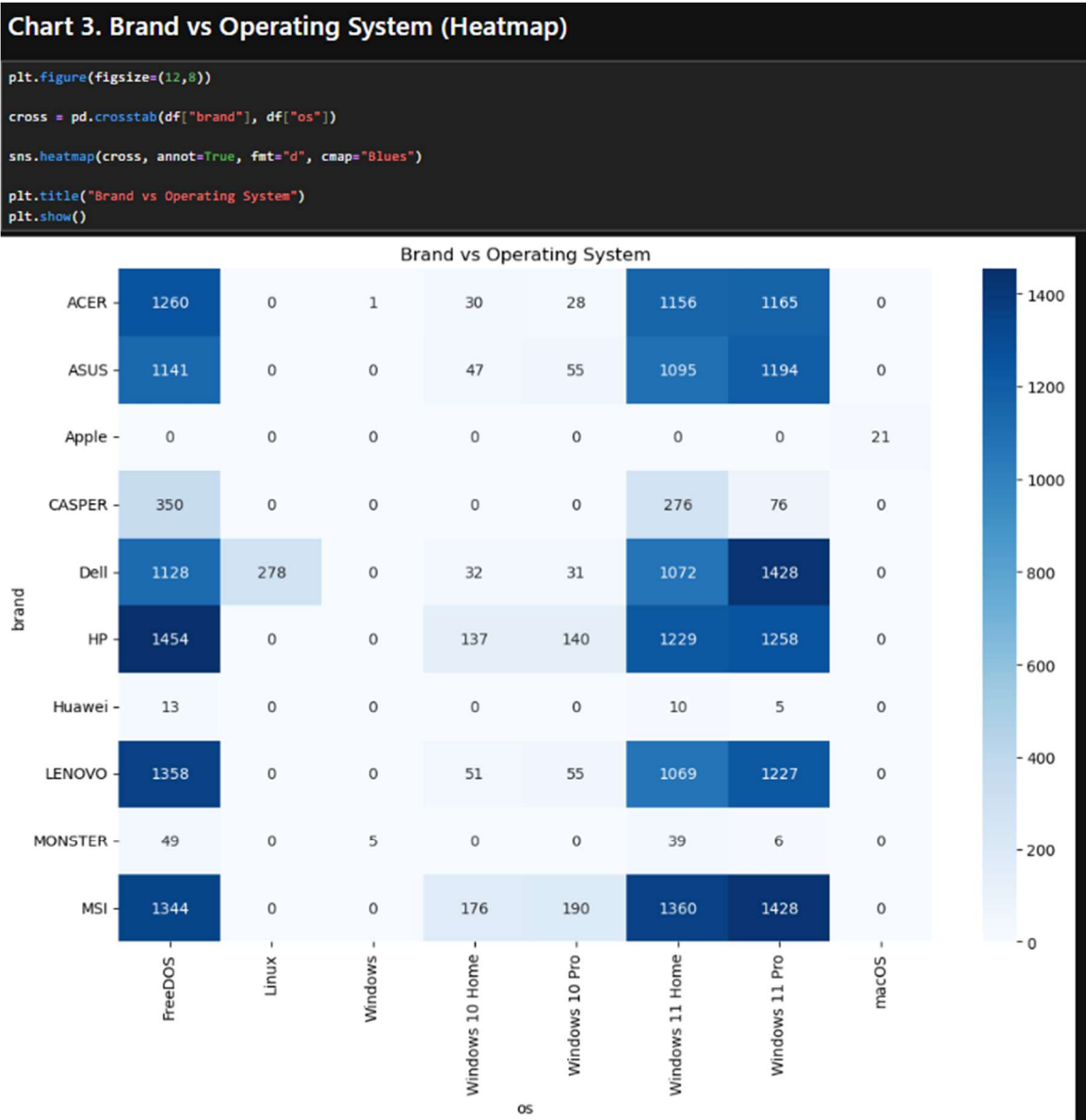


Chart 4: Operating System Distribution (Pie Chart)

Interpretation: The pie chart illustrates the distribution of operating systems in the dataset, highlighting the most popular operating systems among laptop models.

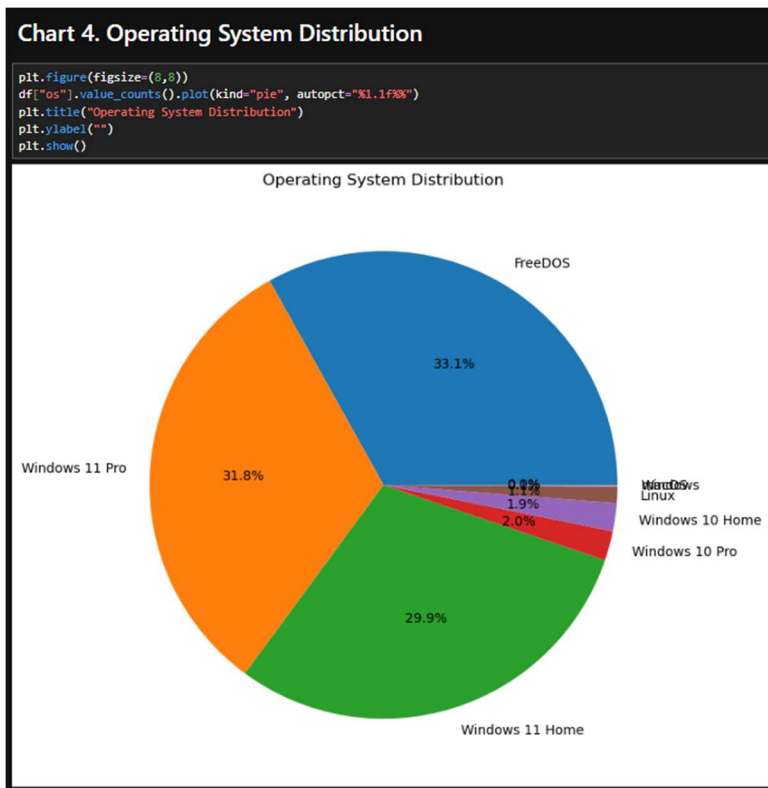


Chart 5: CPU Brand Distribution (Bar Chart)

Interpretation: The bar chart shows the distribution of CPU brands across laptops, helping identify the most commonly used processor brand in the dataset.

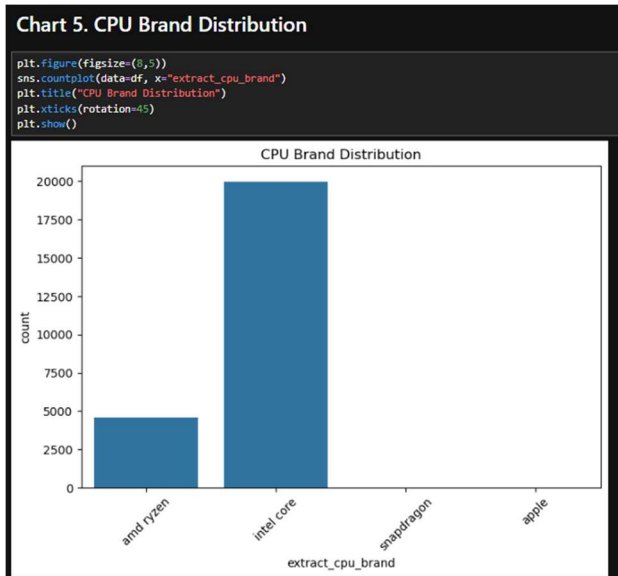


Chart 6: RAM Configuration Distribution (Bar Chart)

Interpretation: The bar chart shows the distribution of different RAM configurations, helping identify the most commonly available RAM sizes in the laptop market.

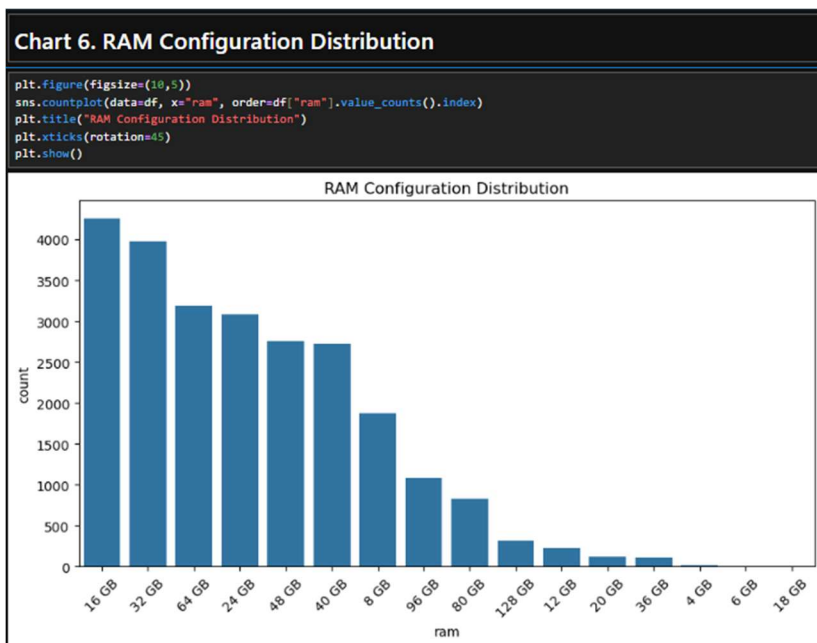


Chart 7: Top 10 GPU Brands (Bar Chart)

Interpretation: The bar chart compares the top 10 most commonly used GPU models, highlighting the graphics cards that dominate the laptop market.

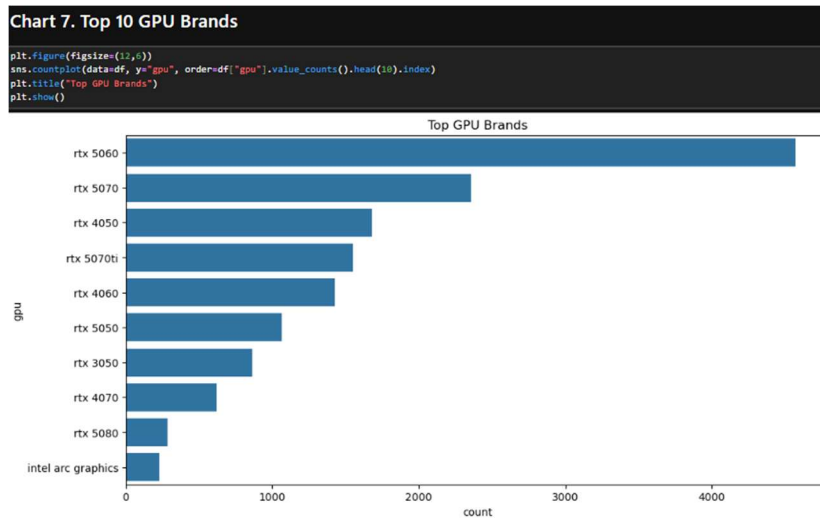


Chart 8: Price Distribution by Brand (Boxplot)

Interpretation: The boxplot compares laptop prices across different brands, helping identify price variations, median prices, and outliers among brands.

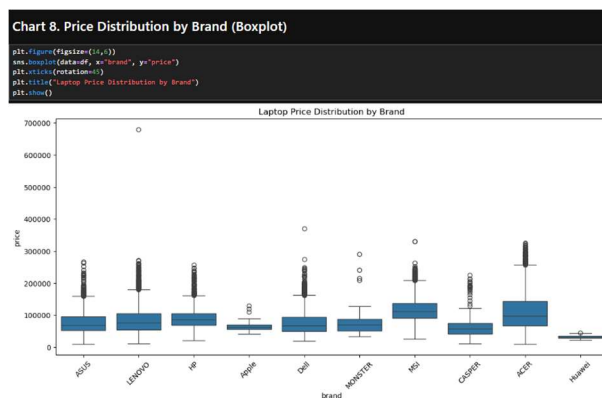


Chart 9: Correlation Heatmap

Interpretation: The correlation heatmap shows the relationship between numerical variables, helping identify which features have the strongest influence on laptop prices.

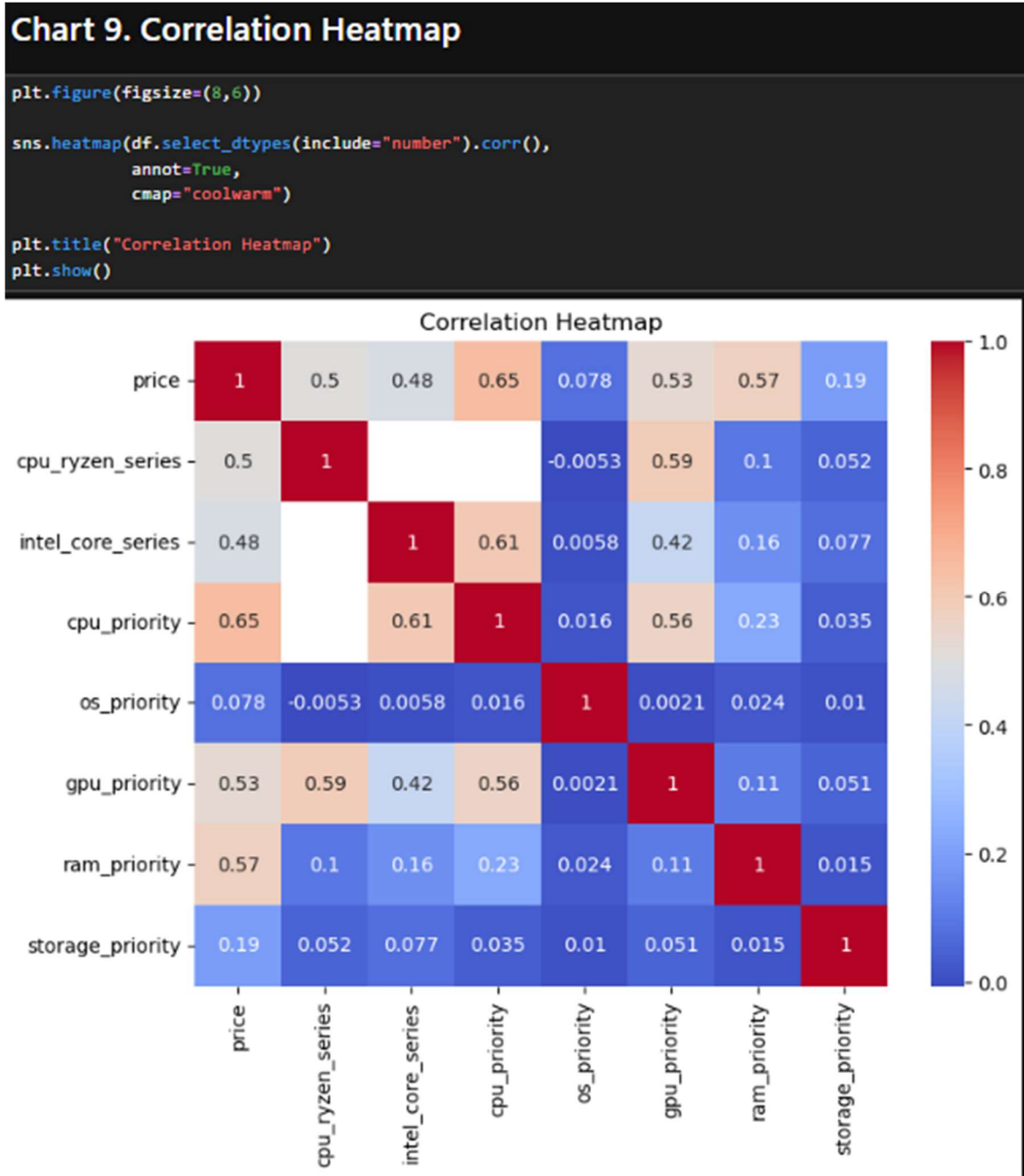


Chart 10: Price vs RAM (Scatter Plot)

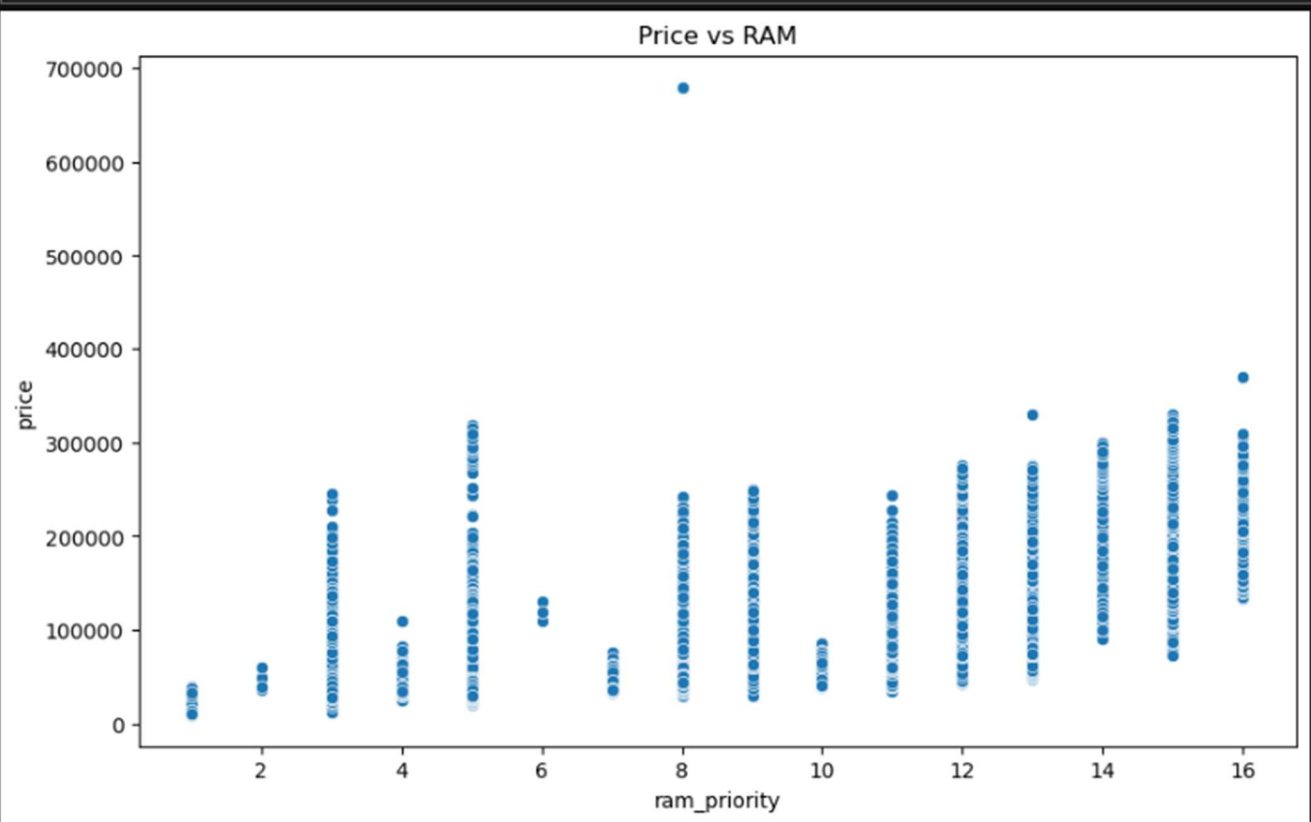
Interpretation: The scatter plot illustrates the relationship between RAM size and laptop price, showing that laptops with higher RAM generally have higher prices.

Chart 10. Price vs RAM (Scatter Plot)

```
plt.figure(figsize=(10,6))

sns.scatterplot(data=df,
                x="ram_priority",
                y="price")

plt.title("Price vs RAM")
plt.show()
```



Conclusion

- This project successfully analyzed the laptop dataset using data cleaning, exploratory data analysis (EDA), and data visualization techniques.
- Successfully cleaned and analyzed the laptop dataset using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn.
- Performed Exploratory Data Analysis (EDA) to identify patterns, trends, and relationships within the dataset.
- Compared different laptop brands based on the number of models, pricing, and product variety.
- Analyzed the impact of hardware specifications such as CPU, GPU, RAM, storage, and operating system on laptop prices.
- Identified the most popular brands, operating systems, RAM configurations, storage options, and usage categories.
- Used various visualizations, including bar charts, pie charts, heatmaps, histograms, boxplots, and scatter plots, to present insights effectively.
- Found that high-performance hardware configurations generally result in higher laptop prices.
- Overall, this project demonstrates how data analytics can transform raw data into meaningful business insights, enabling informed decision-making in the laptop market.